

Steps

Reason

9. $C(c) \wedge L(c)$

Conjunction on 5, 8

10. $\exists x (C(x) \wedge L(x))$

Existential generalization on 9

15) For each of these arguments determine whether the argument is correct or incorrect and explain why.

a) All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic.

Soln: $S(x)$: x is a student in this class $L(x)$: x understands Logic

Premise: 1. $\forall x (S(x) \Rightarrow L(x))$ 2. $S(\text{Xavier})$

Conclusion: $L(\text{Xavier})$ Domain: All humans

Steps

Reason

1. $\forall x (S(x) \Rightarrow L(x))$

Premise
Premise

2. $S(\text{Xavier})$

3. $S(\text{Xavier}) \Rightarrow L(\text{Xavier})$ Universal instantiation on 1

4. $L(\text{Xavier})$ Modus ponens on 2, 3

This is a valid argument since the conclusion is derived using valid rules of inference.

b) Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major.

Soln: $C(x)$: x is a computer science major $DM(x)$: x takes discrete mathematics.

Premise: 1. $\forall x (C(x) \Rightarrow DM(x))$ 2. $DM(\text{Natasha})$

Conclusion: $C(\text{Natasha})$ Domain: All humans